



## COURSE OUTLINE

|    |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                     |                                                                 |                 |
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| 1  | Faculty                      | Faculty of Engineering& Applied Sciences (FEAS)                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                     |                                                                 |                 |
| 2  | Department                   | Department of Mathematics                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                     |                                                                 |                 |
| 3  | Program                      | Bachelor of Science in Computer Science and Engineering (B.Sc. in CSE)                                                                                                                                                                                                                                                                                                                                                                                                                                 |                     |                                                                 |                 |
| 4  | Course Code                  | MAT 221                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                     |                                                                 |                 |
| 5  | Course Title                 | Linear Algebra, Differential Equations and Fourier Analysis                                                                                                                                                                                                                                                                                                                                                                                                                                            |                     |                                                                 |                 |
| 6  | Course Type                  | Basic Math Course                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                     |                                                                 |                 |
| 7  | Pre-requisites               | MAT 101, MAT 111                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                     |                                                                 |                 |
| 8  | Credit Hours/Value           | 3.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                     |                                                                 |                 |
| 9  | Contact Hours                | 17 weeks × 2 classes per week × 1.25 hours per class = 42.5 total hours                                                                                                                                                                                                                                                                                                                                                                                                                                |                     |                                                                 |                 |
| 10 | Year- Semester               | 2 - 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                     |                                                                 |                 |
| 11 | Academic Session             | Spring 2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                     |                                                                 |                 |
| 12 | Class Schedule               | Intake – Section (Shift)                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Class Day           | Class Hours                                                     | Venue           |
|    |                              | 53–1 (Day)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Monday              | 10:30 AM – 11:45 AM                                             | B-2/319         |
|    |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Tuesday             | 02:45 PM – 04:00 PM                                             | B-1/203         |
|    |                              | BUBT Campus, Rupnagar, Mirpur 2, Dhaka - 1216                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                     |                                                                 |                 |
| 13 | Course Website               | Google Classroom Code:<br>Meet Link:                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                     |                                                                 |                 |
| 14 | Course Teacher’s Information | Name (Code): Md. Abdul Hye (ABH)                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                     | Specialization: Mathematical Biology, Fluid Dynamics.           |                 |
|    |                              | Designation: Assistant Professor<br>Department of Mathematics and Statistics                                                                                                                                                                                                                                                                                                                                                                                                                           |                     | Email: <a href="mailto:abdul@bubt.edu.bd">abdul@bubt.edu.bd</a> |                 |
|    |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                     | Cell No. 01913406805                                            | Room: 811 (B-2) |
| 15 | Counselling Hour/Tutorial    | Day                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Counseling Hours    | Venue                                                           |                 |
|    |                              | Monday                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 11:45 AM – 01:00 PM | Room: 811 (B-2)                                                 |                 |
| 16 | Accessories & Aids           | Students must carry learning materials like lecture notes, calculator, pen, pencil, eraser etc. in the classroom. Borrowing learning materials in the classroom or exam room from fellow students is prohibited. A student is also advised to keep a separate class note (khata) of 50 pages for the course during class hours.                                                                                                                                                                        |                     |                                                                 |                 |
| 17 | Course Rationale             | Maximum topics will be covered from the textbook. For the rest of the topics, reference books will be followed. Some class notes will be uploaded on the web. White board will be used for most of the time. Multimedia projector and a PC will be used for the convenience of the students to understand codes practically. Students must participate in classroom discussions for case studies, problems solving and project developments.                                                           |                     |                                                                 |                 |
| 18 | Course Objective             | The course has therefore been designed in a manner<br>Obj. 1. Construct, or give examples of mathematical expressions that involve vectors, matrices, and linear systems of linear equations. Apply linear algebra concepts to model, solve, and analyze real-world situations.<br>Obj. 2. Students should understand the practical importance of solving differential equations.<br>Obj. 3. Students will be able to use Fourier series describe a periodic signal in terms of cosine and sine waves. |                     |                                                                 |                 |
| 19 | Course Learning              | On successful completion of this course, students will be able to                                                                                                                                                                                                                                                                                                                                                                                                                                      |                     |                                                                 |                 |

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| <b>Outcomes (CLOs)</b> | <b>CLO1.</b> Understanding the vectors in $\mathbb{R}^n$ and $\mathbb{C}^n$ , system of linear equations and matrix algebra. including their various operations. (K3 & K4; P1 & P2)                                                  |
|                        | <b>CLO2.</b> Analyzing mathematical statements and expressions (for example, to assess whether a particular statement is accurate, or to describe solutions of systems in terms of existence and uniqueness). (K3 & K4; P1, P2 & P3) |
|                        | <b>CLO3.</b> Understanding and applying differential equations in different fields of engineering with their various solution methods. (K3 & K4; P1, P2, P3 & P7)                                                                    |
|                        | <b>CLO4.</b> Applying the concept of Fourier series, and Fourier transform to solve signal processing. (K3 & K4; P1, P2, P3 & P7)                                                                                                    |

|                                                                          |                                                                                           |                |                    |                            |                                                                                                                                                  |                                                                                                                                                                                        |
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| 20                                                                       | Mapping of CO – PO – K, P/A – Learning Domain, and Method of Delivery and Assessment Tool |                |                    |                            |                                                                                                                                                  |                                                                                                                                                                                        |
|                                                                          | CLOs                                                                                      | PLOs & CF      | Learning Domain    | Ks & Ps**                  | Delivery Methods / Activities                                                                                                                    | Assessment Tools                                                                                                                                                                       |
|                                                                          | CLO1                                                                                      | PLO1<br>CF = 2 | Remembering        | K3& K4;<br>P1& P2          | ⊙ Classroom Lecture (PPP &/or WBT*)<br>⊙ Classroom Discussion& Exercise practice<br>⊙ Analysis and design problem solving<br>⊙ Course Counseling | <b>Indirect:</b> In class response, course counseling, course end survey<br><br><b>Direct:</b> Class participation and activity, class test, assignment midterm and final examinations |
|                                                                          | CLO2                                                                                      | PLO2<br>CF = 3 | Understand & Apply | K3 & K4;<br>P1, P2 & P3    |                                                                                                                                                  |                                                                                                                                                                                        |
|                                                                          | CLO3                                                                                      | PLO2<br>CF = 3 | Remembering        | K3& K4;<br>P1, P2, P3 & P7 |                                                                                                                                                  |                                                                                                                                                                                        |
|                                                                          | CLO4                                                                                      | PLO2<br>CF = 3 | Understand & Apply | K3 & K4;<br>P1, P2, P3& P7 |                                                                                                                                                  |                                                                                                                                                                                        |
| *PPP: Power Point Presentation & WBL: White Board Teaching               |                                                                                           |                |                    |                            |                                                                                                                                                  |                                                                                                                                                                                        |
| **Ks : Knowledge Profiles & Ps : Profiles of Complex Engineering Problem |                                                                                           |                |                    |                            |                                                                                                                                                  |                                                                                                                                                                                        |
| Correlation of COs to POs                                                |                                                                                           |                |                    |                            |                                                                                                                                                  |                                                                                                                                                                                        |
| Correlation Criteria                                                     |                                                                                           |                |                    |                            | Correlation Level                                                                                                                                | Correlation Factor (CF)                                                                                                                                                                |
| Less than 25%                                                            |                                                                                           |                |                    |                            | Almost no correlation                                                                                                                            | 0                                                                                                                                                                                      |
| Equal to or greater than 25% and below50%                                |                                                                                           |                |                    |                            | Poor (Low)                                                                                                                                       | 1                                                                                                                                                                                      |
| Equal to or greater than 50% and below 75%                               |                                                                                           |                |                    |                            | Moderate (Medium)                                                                                                                                | 2                                                                                                                                                                                      |
| Equal to or greater than 75%and up to 100%                               |                                                                                           |                |                    |                            | Significant (High)                                                                                                                               | 3                                                                                                                                                                                      |

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| <b>21 Teaching-Learning Strategy</b> | Maximum topics will be covered from the textbook. For the rest of the topics, reference books will be followed. Some class notes will be uploaded on the web. White board will be used for most of the time. Multimedia projector and a PC will be used for the convenience of the students to understand codes practically. Students must participate in classroom discussions for case studies, problems solving and project developments. |
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| 22 | <b>Course Plan</b>                                                                                                                                                                                                                                                                         |                                                          |                                                                                  |                                      |                                           |                              |             |
|    | This course consists of 150.0 min./week of class contact hours and an additional 90 min./week of counseling hours to explain students’ design problems, provide reading materials, and assist in understanding lecture materials for preparing examinations, class tests, and assignments. |                                                          |                                                                                  |                                      |                                           |                              |             |
|    | Week                                                                                                                                                                                                                                                                                       | Lecture                                                  | Lecture Topic<br>(syllabus covered)                                              | Lecture Note, Text Book & Other Ref. | Teaching-Learning Strategy                | Assessment Strategy          | CO          |
|    | 1                                                                                                                                                                                                                                                                                          | 1                                                        | Introduction to Matrix, Types of matrix, Algebra of matrix, Inverse of a matrix. | Chapter: 2                           | Delivering presentation-based instruction | -                            | CLO1        |
|    |                                                                                                                                                                                                                                                                                            | 2                                                        | Echelon matrix and Canonical Form. Elementary row operations, Rank of Matrix.    | Chapter: 2,3                         | Delivering lecture-based instruction      | Class test 1 & Midterm Exam. | CLO1 & CLO2 |
|    | 2                                                                                                                                                                                                                                                                                          | 3                                                        |                                                                                  |                                      |                                           |                              |             |
| 3  | 4                                                                                                                                                                                                                                                                                          |                                                          |                                                                                  |                                      |                                           |                              |             |
| 3  | 5                                                                                                                                                                                                                                                                                          | Systems of Linear Equations, Use Echelon form in solving |                                                                                  |                                      |                                           |                              |             |

|                                           |                                                 |    |                                                                                                                                                                             |            |                                                                                                                                          |                            |      |  |
|-------------------------------------------|-------------------------------------------------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------|--|
|                                           |                                                 | 6  | system of homogeneous and non-homogeneous equations.                                                                                                                        | Chapter: 4 | Delivering lecture-based instruction, classroom discussion, example problem solving and assign assignment and disclose in a presentation | Assignment & Midterm Exam. | CLO2 |  |
|                                           | 4                                               | 7  | Vector spaces, subspaces,                                                                                                                                                   |            |                                                                                                                                          |                            |      |  |
|                                           |                                                 | 8  | Linear combination of vectors.                                                                                                                                              |            |                                                                                                                                          |                            |      |  |
|                                           | 5                                               | 9  | Linear independent & dependent of vector, linear,                                                                                                                           |            |                                                                                                                                          |                            |      |  |
|                                           |                                                 | 10 | basis & dimension of vectors.                                                                                                                                               |            |                                                                                                                                          |                            |      |  |
|                                           | 6                                               | 11 | Eigenvalues and Eigenvectors.                                                                                                                                               | Chapter: 9 | Delivering lecture-based instruction, classroom discussion and example problem solving                                                   | Class test 1               | CLO2 |  |
|                                           |                                                 | 12 | Diagonalizations, Cayley-Hamilton Theorem.                                                                                                                                  |            |                                                                                                                                          |                            |      |  |
|                                           | 7                                               | 13 | Introduction to differential equations, Classification of Differential equations, Order and degree of a Differential Equation. Formation of ordinary differential equation. | Chapter: 1 | Delivering lecture-based instruction, classroom discussion, example problem solving and assign assignment and disclose in a presentation | Assignment & Midterm Exam. | CLO3 |  |
|                                           |                                                 | 14 |                                                                                                                                                                             |            |                                                                                                                                          |                            |      |  |
|                                           | 8                                               | 15 | Solutions of ordinary differential equations by the method variable separable,                                                                                              | Chapter 2  | Delivering lecture-based instruction, classroom discussion and example problem solving                                                   | Midterm Exam.              |      |  |
|                                           |                                                 | 16 | Solutions of homogeneous ordinary differential equations.                                                                                                                   |            |                                                                                                                                          |                            |      |  |
|                                           | Midterm Examination (10 March - 25 March, 2025) |    |                                                                                                                                                                             |            |                                                                                                                                          |                            |      |  |
|                                           | 9                                               | 17 | Solutions Exact differential equations. Integrating factor and solutions of non-exact DE.                                                                                   | Chapter 2  | Delivering lecture-based instruction, classroom discussion, example problem solving and assign assignment and disclose in a presentation | Assignment Final Exam.     | CLO3 |  |
|                                           |                                                 | 18 |                                                                                                                                                                             |            |                                                                                                                                          |                            |      |  |
|                                           | 10                                              | 19 | Linear differential Equation, Bernoulli Equations and Special Integrating Factor (IF), orthogonal trajectories.                                                             | Chapter 2  |                                                                                                                                          |                            |      |  |
|                                           |                                                 | 20 |                                                                                                                                                                             |            |                                                                                                                                          |                            |      |  |
|                                           | 11                                              | 21 | Homogeneous Linear differential equation with constant coefficient.                                                                                                         | Chapter 4  | Delivering lecture-based instruction, classroom discussion and example problem solving                                                   | Class Test 2 Final Exam.   | CLO3 |  |
|                                           |                                                 | 22 |                                                                                                                                                                             |            |                                                                                                                                          |                            |      |  |
|                                           | 12                                              | 23 | Non-homogeneous Linear differential equation with constant coefficients.                                                                                                    |            |                                                                                                                                          |                            |      |  |
|                                           |                                                 | 24 |                                                                                                                                                                             |            |                                                                                                                                          |                            |      |  |
|                                           | 13                                              | 25 | Undeterminant coefficients, variation of parameters, Cauchy Euler equation.                                                                                                 | Chapter 3  | Assignment submission with a presentation                                                                                                | Presentation               | CLO4 |  |
|                                           |                                                 | 26 |                                                                                                                                                                             |            |                                                                                                                                          |                            |      |  |
|                                           | 14                                              | 27 | Modeling with first order differential equation.                                                                                                                            |            |                                                                                                                                          |                            |      |  |
|                                           |                                                 | 28 |                                                                                                                                                                             |            |                                                                                                                                          |                            |      |  |
|                                           | 15                                              | 29 | System of linear differential equation, Non-linear differential equation, Partial differential equation.                                                                    | Chapter 4  | Delivering lecture-based instruction, classroom discussion and example problem solving                                                   | Final Exam.                |      |  |
|                                           |                                                 | 30 |                                                                                                                                                                             |            |                                                                                                                                          |                            |      |  |
|                                           | 16                                              | 31 | Periodic function, Dirichlet conditions, Fourier Series, even and odd function.                                                                                             | Chapter 2  | Delivering lecture-based instruction, classroom discussion and example problem solving                                                   |                            |      |  |
|                                           |                                                 | 32 |                                                                                                                                                                             |            |                                                                                                                                          |                            |      |  |
|                                           | 17                                              | 33 | Half range Fourier sine or cosine series, uniform convergence, complex form of Fourier series. Fourier Integral, Fourier Transform                                          |            |                                                                                                                                          |                            |      |  |
|                                           |                                                 | 34 |                                                                                                                                                                             |            |                                                                                                                                          |                            |      |  |
| Final Examination (18 June-02 July, 2025) |                                                 |    |                                                                                                                                                                             |            |                                                                                                                                          |                            |      |  |

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| 23 i | Class note | 1. Class Note of the course of Linear Algebra, Differential Equations, Fourier Analysis for the students of B.Sc. in CSE Program, BUBT by Rama Sarkar. |
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| ii                                                  | Text Books                                    | <div>2. Md. Abdur Rahman-Linea Algebra (2018).</div> <div>3. (Schaum’s Outlines) Seymour Lipschutz, Marc Lars Lipson - Linear Algebra-McGraw-Hill (2009).</div> <div>4. Md. Abu Yousuf-Ordinary Differential Equation (2018).</div> <div>5. Dennis G. Zill - A First Course in Differential Equations with Modeling Applications-Brooks Cole (2008).</div> <div>6. Murray R. Spiegel - Schaum's Outline of Fourier Analysis with Applications to Boundary Value Problems -McGraw-Hill (1974).</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |               |          |                 |                   |                       |                                  |    |    |            |   |                     |                                     |   |                  |                     |   |    |                   |   |    |            |  |     |                                                     |  |  |                   |                       |  |           |    |  |           |    |  |           |    |  |           |    |  |           |    |  |               |                                               |  |
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| 24                                                  | Assessment and marks distribution criteria    | <div>Active engagement in class activities, participation in outside classroom discussions, and communication through the Internet and phone are integral parts of this course. Failure to participate in class regularly, take class tests, and/or complete assignments may result in failing the course. To achieve the course-specific expectations, students must actively participate in classroom discussions and complete all sets of work at a satisfactory level, as outlined in the course content. The course-specific expectations for students are achieved if</div> <div><div><div>75% of students in a section attend more than 70% of classes (determined by summative assessment).</div><div>Their active participation in the classroom discussion is targeted at up to 80% of the total attendees (determined by formative assessments such as eye contact, clicker questions, and group discussions).</div><div>Equal or more than 40% of course outcomes must be achieved by the students (summative assessment).</div><div>The level of engagement in the studies, such as regularly preparing class lectures, class tests, and assignments, must be more than 60% (formative assessment).</div><div>Expected level of participation in the outside class discussion (once weak, more than 30% of students in a section) by course counseling and using social media like Google Classroom, email, phone call, etc. (formative assessment).</div><div>Students are assessed according to their individual performance in the examinations, class tests, assignments, and class participation. The final mark calculation and course outcome assessment are done based on the following mark distribution criteria:</div></div><table><tr><th>Assessment tool</th><th>Conducting Number</th><th>Mark distribution (%)</th></tr><tr><td>Class participation and activity</td><td>34</td><td>05</td></tr><tr><td>Class test</td><td>2</td><td><math>7.5 \times 2 = 15</math></td></tr><tr><td>Assignments/Report and Presentation</td><td>1</td><td><math>5.0 + 5.0 = 10</math></td></tr><tr><td>Midterm examination</td><td>1</td><td>30</td></tr><tr><td>Final examination</td><td>1</td><td>40</td></tr><tr><td colspan="2">Total Mark</td><td>100</td></tr><tr><td colspan="3">Class participation &amp; activity performance criteria</td></tr><tr><th>Performance level</th><th colspan="2">Mark distribution (%)</th></tr><tr><td>90% -100%</td><td colspan="2">05</td></tr><tr><td>85% - 89%</td><td colspan="2">04</td></tr><tr><td>80% - 84%</td><td colspan="2">03</td></tr><tr><td>75% - 89%</td><td colspan="2">02</td></tr><tr><td>70% - 74%</td><td colspan="2">01</td></tr><tr><td>less than 50%</td><td colspan="2">Not allowed to sit for the final examination.</td></tr></table></div> |               |          | Assessment tool | Conducting Number | Mark distribution (%) | Class participation and activity | 34 | 05 | Class test | 2 | $7.5 \times 2 = 15$ | Assignments/Report and Presentation | 1 | $5.0 + 5.0 = 10$ | Midterm examination | 1 | 30 | Final examination | 1 | 40 | Total Mark |  | 100 | Class participation & activity performance criteria |  |  | Performance level | Mark distribution (%) |  | 90% -100% | 05 |  | 85% - 89% | 04 |  | 80% - 84% | 03 |  | 75% - 89% | 02 |  | 70% - 74% | 01 |  | less than 50% | Not allowed to sit for the final examination. |  |
| Assessment tool                                     | Conducting Number                             | Mark distribution (%)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |               |          |                 |                   |                       |                                  |    |    |            |   |                     |                                     |   |                  |                     |   |    |                   |   |    |            |  |     |                                                     |  |  |                   |                       |  |           |    |  |           |    |  |           |    |  |           |    |  |           |    |  |               |                                               |  |
| Class participation and activity                    | 34                                            | 05                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |          |                 |                   |                       |                                  |    |    |            |   |                     |                                     |   |                  |                     |   |    |                   |   |    |            |  |     |                                                     |  |  |                   |                       |  |           |    |  |           |    |  |           |    |  |           |    |  |           |    |  |               |                                               |  |
| Class test                                          | 2                                             | $7.5 \times 2 = 15$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |               |          |                 |                   |                       |                                  |    |    |            |   |                     |                                     |   |                  |                     |   |    |                   |   |    |            |  |     |                                                     |  |  |                   |                       |  |           |    |  |           |    |  |           |    |  |           |    |  |           |    |  |               |                                               |  |
| Assignments/Report and Presentation                 | 1                                             | $5.0 + 5.0 = 10$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |               |          |                 |                   |                       |                                  |    |    |            |   |                     |                                     |   |                  |                     |   |    |                   |   |    |            |  |     |                                                     |  |  |                   |                       |  |           |    |  |           |    |  |           |    |  |           |    |  |           |    |  |               |                                               |  |
| Midterm examination                                 | 1                                             | 30                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |          |                 |                   |                       |                                  |    |    |            |   |                     |                                     |   |                  |                     |   |    |                   |   |    |            |  |     |                                                     |  |  |                   |                       |  |           |    |  |           |    |  |           |    |  |           |    |  |           |    |  |               |                                               |  |
| Final examination                                   | 1                                             | 40                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |          |                 |                   |                       |                                  |    |    |            |   |                     |                                     |   |                  |                     |   |    |                   |   |    |            |  |     |                                                     |  |  |                   |                       |  |           |    |  |           |    |  |           |    |  |           |    |  |           |    |  |               |                                               |  |
| Total Mark                                          |                                               | 100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |               |          |                 |                   |                       |                                  |    |    |            |   |                     |                                     |   |                  |                     |   |    |                   |   |    |            |  |     |                                                     |  |  |                   |                       |  |           |    |  |           |    |  |           |    |  |           |    |  |           |    |  |               |                                               |  |
| Class participation & activity performance criteria |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |          |                 |                   |                       |                                  |    |    |            |   |                     |                                     |   |                  |                     |   |    |                   |   |    |            |  |     |                                                     |  |  |                   |                       |  |           |    |  |           |    |  |           |    |  |           |    |  |           |    |  |               |                                               |  |
| Performance level                                   | Mark distribution (%)                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |          |                 |                   |                       |                                  |    |    |            |   |                     |                                     |   |                  |                     |   |    |                   |   |    |            |  |     |                                                     |  |  |                   |                       |  |           |    |  |           |    |  |           |    |  |           |    |  |           |    |  |               |                                               |  |
| 90% -100%                                           | 05                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |          |                 |                   |                       |                                  |    |    |            |   |                     |                                     |   |                  |                     |   |    |                   |   |    |            |  |     |                                                     |  |  |                   |                       |  |           |    |  |           |    |  |           |    |  |           |    |  |           |    |  |               |                                               |  |
| 85% - 89%                                           | 04                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |          |                 |                   |                       |                                  |    |    |            |   |                     |                                     |   |                  |                     |   |    |                   |   |    |            |  |     |                                                     |  |  |                   |                       |  |           |    |  |           |    |  |           |    |  |           |    |  |           |    |  |               |                                               |  |
| 80% - 84%                                           | 03                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |          |                 |                   |                       |                                  |    |    |            |   |                     |                                     |   |                  |                     |   |    |                   |   |    |            |  |     |                                                     |  |  |                   |                       |  |           |    |  |           |    |  |           |    |  |           |    |  |           |    |  |               |                                               |  |
| 75% - 89%                                           | 02                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |          |                 |                   |                       |                                  |    |    |            |   |                     |                                     |   |                  |                     |   |    |                   |   |    |            |  |     |                                                     |  |  |                   |                       |  |           |    |  |           |    |  |           |    |  |           |    |  |           |    |  |               |                                               |  |
| 70% - 74%                                           | 01                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |          |                 |                   |                       |                                  |    |    |            |   |                     |                                     |   |                  |                     |   |    |                   |   |    |            |  |     |                                                     |  |  |                   |                       |  |           |    |  |           |    |  |           |    |  |           |    |  |           |    |  |               |                                               |  |
| less than 50%                                       | Not allowed to sit for the final examination. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |          |                 |                   |                       |                                  |    |    |            |   |                     |                                     |   |                  |                     |   |    |                   |   |    |            |  |     |                                                     |  |  |                   |                       |  |           |    |  |           |    |  |           |    |  |           |    |  |           |    |  |               |                                               |  |
| 25                                                  | Assessment Strategy                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |          |                 |                   |                       |                                  |    |    |            |   |                     |                                     |   |                  |                     |   |    |                   |   |    |            |  |     |                                                     |  |  |                   |                       |  |           |    |  |           |    |  |           |    |  |           |    |  |           |    |  |               |                                               |  |
|                                                     | Assessment tool                               | Content, Length and Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Weight (mark) | Due date |                 |                   |                       |                                  |    |    |            |   |                     |                                     |   |                  |                     |   |    |                   |   |    |            |  |     |                                                     |  |  |                   |                       |  |           |    |  |           |    |  |           |    |  |           |    |  |           |    |  |               |                                               |  |

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|  | Class test 1                   | Composed of 4 to 6 short questions which cover the topic of design of flat plates and columns under uniaxial loading. Duration of the class test is 40 minutes, and the assigned full mark is 7.5.                                                                                                                                                                                            | 7.5  | On the 12th lecture                                                                                        |
|  | Class test 2                   | Composed of 4 to 6 short questions which cover the topics of structural design of footings. Duration of the class test is 40 minutes, and the assigned full mark is 7.5.                                                                                                                                                                                                                      | 7.5  | On the 25th lecture                                                                                        |
|  | Assignment with a presentation | Full length design of flat slabs using Direct Design Method, columns under biaxial bending and RCC shear wall subjected to axial load and flexure, and seismic detailing of columns and beams are assigned to each student. Assignment with a presentation will be delivered two weeks after assigning. 5 marks are assigned for the assignment and the other 5 marks are for a presentation. | 10.0 | Presentation will be conducted on the 28th lecture                                                         |
|  | Midterm Examination            | Midterm Examination is held according to the Academic Calendar and examination schedule after 8 weeks of classes. All three sets of questions are to be answered by the students in one and half-hour exam period. Each question set contains multiple short type questions. The assigned full mark for the 3 sets of questions is 30.                                                        | 30   | After 16 <sup>th</sup> lecture and following the examination schedule of BUBT.                             |
|  | Final Examination              | Final Examination is held after 8 weeks of classes after the midterm examination. All four sets of questions are to be answered by the students in a two-hour exam period. Each question set contains multiple short type questions. The assigned full mark for the 4 sets of question is 40.                                                                                                 | 40   | After 9 weeks of classes afterward the midterm examination and following the examination schedule of BUBT. |

| 26 | CLO Assessment Criteria | Assessment of CLOs       |       |       |       |       |                 |
|----|-------------------------|--------------------------|-------|-------|-------|-------|-----------------|
|    |                         | Assessment Tool          | CLO   |       |       |       | Mark Allocation |
|    |                         |                          | CLO1  | CLO2  | CLO3  | CLO4  |                 |
|    |                         | Class Participation      | 5/4   | 5/4   | 5/4   | 5/4   | 05              |
|    |                         | Class Test               | 3.5   | 4     | 3.5   | 4     | 15              |
|    |                         | Assignment& Presentation | 0     | 5.0   | 5.0   | 0     | 10              |
|    |                         | Midterm Exam             | 10    | 20    | 0     | 0     | 30              |
|    |                         | Final Exam               | 0     | 0     | 20    | 20    | 40              |
|    |                         | Total Mark               | 14.75 | 30.25 | 29.75 | 25.25 | 100             |

| 27 | Rubrics (Attainment Criteria) | CLOs (Taxonomy domain)                                                                          | Not attained /Failed (0-39%)                                                                                                    | Poor (40%-49%)                                                                                                                                                                                                     | Moderate (50%-64%)                                                                                                                | Good (65%-79%)                                                                                                                                      | Excellent (80%-100%)                                                                                                                                 |
|----|-------------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |                               | CLO1 (understand)<br>CLO2 (understand and apply)<br>CLO3 (apply)<br>CLO4 (understand and apply) | The question was answered with serious deficiencies in understanding and explanation. Applicable method was not almost touched. | The question was answered inadequately by touching on the applicable method or without explanations. As a result, a few steps of problem-solving procedures or concepts are not developed properly or are missing. | The question was answered partially correctly by applying the method or concepts asked, but a few important details were missing. | The question was answered correctly but briefly, and missed some portions of the important explanation by applying the required method or concepts. | The question was answered correctly with detailed explanations using the asking method of solving the problem or concepts with adequate explanation. |

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| 28 | Feedback | All kinds of feedback to the students will be produced within a week after the day of holding a class test and midterm examination. No answer script will be shown for the |
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|  |  | final examination if it is not challenged by a student. Online and email queries will also be responded to within three days by email. |
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| 29 | <b>Grading Policy</b> | Letter grades and grade points are used to evaluate the performance of a student in the course: |                              |                    |
|    |                       | <b>Marks Range</b>                                                                              | <b>Letter Grade</b>          | <b>Grade Point</b> |
|    |                       | 80% and above                                                                                   | A+ : A Plus (Outstanding)    | 4.00               |
|    |                       | 75% to less than 80%                                                                            | A : A regular (Excellent)    | 3.75               |
|    |                       | 70% to less than 75%                                                                            | A- : A minus (Very good)     | 3.50               |
|    |                       | 65% to less than 70%                                                                            | B+ : B Plus (Good)           | 3.25               |
|    |                       | 60% to less than 65%                                                                            | B : B regular (Average)      | 3.00               |
|    |                       | 55% to less than 60%                                                                            | B- : B minus (Below average) | 2.75               |
|    |                       | 50% to less than 55%                                                                            | C+ : C Plus (Poor)           | 2.50               |
|    |                       | 45% to less than 50%                                                                            | C : C regular (Very poor)    | 2.25               |
|    |                       | 40% to less than 45%                                                                            | D : Pass marginally          | 2.00               |
|    |                       | Less than 40%                                                                                   | F : Fail                     | 0.00               |
|    |                       |                                                                                                 | I : Incomplete               |                    |
|    |                       |                                                                                                 | R : Retake                   |                    |
|    |                       |                                                                                                 | W : Withdraw                 |                    |

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| 30 | <b>Additional Course Policies</b> | Assignment              | <p>Late assignment submissions will not be accepted.</p> <p><b><i>Any kind of copied information without proper citation (i.e., plagiarism) in the assignments or any other work is prohibited and treated as serious academic misconduct, which is prosecuted through the Discipline Committee of BUBT. All copied or plagiarized assignments/reports/test papers will be canceled, and the isolated student must prepare a new assignment/project again. Zero tolerance will be shown in this regard.</i></b></p> <p>Feedback after the submission of long assignments will be provided on hand or by email within two weeks.</p> |
|    |                                   | Class Test              | Two class tests (CTs) will be conducted for the course. All class tests have equal weightage of 7.5. Both regular and surprise CTs can be conducted.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|    |                                   | Closed book assessments | CTs, midterm and final examinations are closed book assessments. Mobile phone is prohibited in the examination hall. Students are insisted on carrying simple scientific calculators to solve the complex calculations and a wrist-watch to follow time during the exam hours.                                                                                                                                                                                                                                                                                                                                                      |
|    |                                   | Test Policy             | If a student is absent from a class test anyway and does not report to the class teacher personally beforehand, his/her score for that test will be zero. No make-up for the class test will be allowed. No supplementary for midterm and final examinations will be entertained without physical presence and recommendation of the guardian, along with written permission of the department. Supplementary examination questions are much harder than the regular examination questions; therefore, students are discouraged from taking supplementary examinations.                                                             |

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| 31 | <b>Additional Information</b> | <p>Academic Calendar Fall2024: <a href="https://www.bubt.edu.bd/Home/page_details/Academic_Calender">https://www.bubt.edu.bd/Home/page_details/Academic_Calender</a></p> <p>Academic Rules: <a href="https://www.bubt.edu.bd/Home/page_details/Rules_and_Regulations">https://www.bubt.edu.bd/Home/page_details/Rules_and_Regulations</a></p> <p>Grading &amp; Evaluation: <a href="https://www.bubt.edu.bd/Home/page_details/Evaluation_Grading_System">https://www.bubt.edu.bd/Home/page_details/Evaluation_Grading_System</a></p> <p>Rules&amp; Regulations: <a href="https://www.bubt.edu.bd/Home/page_details/Office_of_the_Registrar">https://www.bubt.edu.bd/Home/page_details/Office_of_the_Registrar</a></p> |
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| 32 | <b>Bloom's Taxonomy for Teaching-Learning</b> | <p>Bloom's Taxonomy is a set of three hierarchical models used to classify educational learning objectives into levels of complexity and specific quality. The three taxonomy domains for achieving learning objectives are cognitive, affective, and psychomotor. Cognitive domain is in the primary focus of educating and frequently used to structure</p> |
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| curriculum learning objectives and achieve the level of learning. The three domains and their respective levels are illustrated below: |                                         |                                          |
| <b>Cognitive [C]</b><br>(Knowledge-based)                                                                                              | <b>Affective [A]</b><br>(Emotion-based) | <b>Psychomotor [P]</b><br>(Action-based) |
| Remember                                                                                                                               | Receive                                 | Imitate                                  |
| Understand                                                                                                                             | Respond                                 | Manipulate                               |
| Apply                                                                                                                                  | Value                                   | Precision                                |
| Analyze                                                                                                                                | Organize                                | Articulation                             |
| Evaluate                                                                                                                               | Characterize                            | Naturalization                           |
| Create                                                                                                                                 | --- --- ---                             | --- --- ---                              |

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| <b>33 Descriptions of Cognitive Domain (Anderson and Krathwohl's updated Taxonomy in 2001):</b>             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                  |
| The <b>cognitive domain</b> involves the development of our mental skills and the acquisition of knowledge. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                  |
| <b>Category (Level)</b>                                                                                     | <b>Meaning</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Keywords</b>                                                                                                                                                                                  |
| Remember (C1)                                                                                               | Recognizing or recalling knowledge from memory. Remembering is when memory is used to produce or retrieve definitions, facts, or lists, or to recite previously learned information.                                                                                                                                                                                                                                                                                                                   | Define, describe, draw, find, identify, label, list, match, name, quote, recall, recite, tell, and write                                                                                         |
| Understand (C2)                                                                                             | Constructing meaning from different types of functions be they written or graphic messages or activities like interpreting, exemplifying, classifying, summarizing, inferring, comparing, or explaining.                                                                                                                                                                                                                                                                                               | Classify, compare, exemplify, conclude, demonstrate, discuss, explain, identify, illustrate, interpret, paraphrase, predict, and report                                                          |
| Apply (C3)                                                                                                  | Carrying out or using a procedure through executing or implementing. Applying relates to or refers to situations where learned material is used through products like models, presentations, interviews, or simulations.                                                                                                                                                                                                                                                                               | Apply, change, choose, compute, dramatize, implement, interview, prepare, produce, role play, select, show, transfer, and use                                                                    |
| Analyze (C4)                                                                                                | Breaking materials or concepts into parts, determining how the parts relate to one another or how they interrelate, or how the parts relate to an overall structure or purpose. Mental actions included in this function are differentiating, organizing, and attributing, as well as being able to distinguish between the components or parts. When one is analyzing, he/she can illustrate this mental function by creating spreadsheets, surveys, charts, or diagrams, or graphic representations. | Analyze, characterize, classify, compare, contrast, debate, deconstruct, deduce, differentiate, discriminate, distinguish, examine, organize, outline, relate, research, separate, and structure |
| Evaluate (C5)                                                                                               | Making judgments based on criteria and standards through checking and critiquing. Critiques, recommendations, and reports are some of the products that can be created to demonstrate the processes of evaluation.                                                                                                                                                                                                                                                                                     | Appraise, argue, assess, choose, conclude, decide, evaluate, judge, justify, predict, prioritize, prove, rank, rate, select, Monitor.                                                            |
| Create (C6)                                                                                                 | Putting elements together to form a coherent or functional whole; reorganizing elements into a new pattern or structure through generating, planning, or producing. Creating requires users to put parts together in a new way, or synthesize parts into something new and different creating a new form or product. This process is the most difficult mental function in the new taxonomy.                                                                                                           | Create, invent, compose, predict, plan, construct, design, propose, devise, and formulate                                                                                                        |

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| <b>34 Graduate Attributes (Program Learning Outcomes) of B.Sc. in CE Program based on Washington Accord</b>                                                                                                                                                                                                                                                                                                                |  |
| Program Learning Outcomes (PLOs) are brief statements that describe what students are expected to know and be able to do by the time of graduation. These relate to the knowledge, skills, and attitudes that students acquire throughout the entire course of a program. The students of the B.Sc. in CE program are expected to achieve the following graduate attributes or program outcomes at the time of graduation: |  |
| <b>PLO1.</b> Apply knowledge of mathematics, natural science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.<br><b>PLO2.</b> Identify, formulate, and analyze complex engineering problems reaching substantiated conclusions using the first principles of mathematics and natural and engineering sciences.                                                |  |

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|  | <p><b>PLO3.</b> Design solutions for complex engineering problems and design systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.</p> <p><b>PLO4.</b> Conduct investigations of complex problems using research-based knowledge and research methods that include the design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.</p> <p><b>PLO5.</b> Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, to complex engineering problems, with an understanding of the limitations.</p> <p><b>PLO6.</b> Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems.</p> <p><b>PLO7.</b> Understand and evaluate the sustainability and impact of professional engineering work to solve complex engineering problems in societal and environmental contexts.</p> <p><b>PLO8.</b> Apply ethical principles and commit to professional ethics, responsibilities, and norms of engineering practice.</p> <p><b>PLO9.</b> Function effectively as individuals and members or leaders of diverse teams and in multidisciplinary settings.</p> <p><b>PLO10.</b> Communicate effectively on complex engineering activities with the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and convey and receive clear instructions.</p> <p><b>PLO11.</b> Demonstrate knowledge and understanding of engineering and management principles and apply them to their work as team members or leaders or entrepreneurs to manage projects in multidisciplinary environments.</p> <p><b>PLO12.</b> Recognize self-awareness to engage in a lifelong learning process to reflect in the broadest context of technological change.</p> |
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| <b>35</b> | <b>Knowledge Profile (K)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|           | <p><b>K1:</b> A systematic, theory-based understanding of the natural sciences applicable to the discipline.</p> <p><b>K2:</b> Conceptually-based mathematics, numerical analysis, statistics and formal aspects of computer and information science to support analysis and modeling applicable to the discipline.</p> <p><b>K3:</b> A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.</p> <p><b>K4:</b> Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline much is at the forefront of the discipline.</p> <p><b>K5:</b> Knowledge that supports engineering design in a practice area.</p> <p><b>K6:</b> Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.</p> <p><b>K7:</b> Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the professional responsibility of an engineer to public safety; the impacts of engineering activity: economic, social, cultural, environmental and sustainability.</p> <p><b>K8:</b> Engagement with selected knowledge in the research literature of the discipline.</p> |

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| <b>36</b> | <b>Range of Complex Engineering Problem Solving (P)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|           | <p>Complex Engineering Problems have characteristic <b>P1</b> and several or all of <b>P2</b> to <b>P7</b>:</p> <p><b>P1. Depth of knowledge required:</b> Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8, which allows a fundamentals-based, first principles analytical approach</p> <p><b>P2. Range of conflicting requirements:</b> Involve wide-ranging or conflicting technical, engineering and other issues</p> <p><b>P3. Depth of analysis required:</b> Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models</p> <p><b>P4. Familiarity of issues:</b> Involve infrequently encountered issues</p> <p><b>P5. Extent of applicable codes:</b> Are outside problems encompassed by standards and codes of practice for professional engineering</p> <p><b>P6. Extent of stakeholder involvement and conflicting requirements:</b> Involve diverse groups of stakeholders with widely varying needs</p> <p><b>P7. Interdependence:</b> Are high-level problems including many component parts or sub-problems</p> |



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| 37 | Range of Complex Engineering Activities (A)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                       |
|    | <p>Attribute Complex activities means (engineering) activities or projects that have several or all of the following characteristics:</p> <p><b>A1. Range of resources:</b> Involve the use of diverse resources (and for this purpose resources include people, money, equipment, materials, information and technologies).</p> <p><b>A2. Level of interaction:</b> Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering, or other issues.</p> <p><b>A3. Innovation:</b> Involve creative use of engineering principles and research-based knowledge in novel ways.</p> <p><b>A4. Consequences for society and the environment:</b> Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation.</p> <p><b>A5. Familiarity:</b> Can extend beyond previous experiences by applying principles-based approaches.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                       |
| 38 | Code of Conduct                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                       |
|    | <p>It is strongly suggested that students keep discipline in the classroom by attending class on time, listening to lectures attentively, and participating in discussions on the subject. To get class participation grades, students <b>MUST</b> attend the classes of the courses s/he registered for. Turn off his or her cell phone before entering a class or participating in class tests and exams. There are activities that are considered academic misconduct. One of them is plagiarism, which signifies the deliberate formal presentation or submission of works, phrases, texts, ideas, illustrations, or diagrams of others as one’s own without proper citation. Another one is the use of unauthorized aids (including electronic devices), asking for assistance, or using illegal materials when preparing assignments or in examinations. In addition, copying from others’ work, showing your work to others, and asking for answers are also considered academic misconduct. Penalties for involving academic misconduct include one or more of the following: a zero grade on the work produced, a failing grade in the course, suspension for one semester or more, and even expulsion from the university. On the university premises or at a university-sponsored program, students must abide by the Student Code of Conduct and other Rules and Regulations of BUBT, which are available on the BUBT website at <a href="https://www.bubt.edu.bd/Home/page_details/Office_of_the_Proctor">https://www.bubt.edu.bd/Home/page_details/Office_of_the_Proctor</a>.</p> |                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                       |
| 39 | Social & Moral Values                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                       |
|    | <p>Our promises are based on the three cardinal principles:</p> <p>i. What we do believe.</p> <p>ii. What we do practice.</p> <p>iii. What we will promote.</p> <p>However, students are advised to undertake the following commitments for social and moral developments.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                       |
|    | <ul style="list-style-type: none"><li>• To be punctual and attentive in classes;</li><li>• To prioritize honesty &amp; faith;</li><li>• To ensure mutual respect;</li><li>• To be always proactive;</li><li>• To avoid conspiracy;</li><li>• To be cooperative in learning;</li><li>• To be sincere in class preparation;</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <ul style="list-style-type: none"><li>• To avoid unfair means and plagiarism in exams, report writings and assignments;</li><li>• To carry out assignments or keep other commitments timely;</li><li>• To be motivated for asking question and encourage feedback;</li><li>• Not to forget to switch-off the cellphone in a class;</li></ul> | <ul style="list-style-type: none"><li>• To follow the dress code and wearing ID card on campus;</li><li>• To be decent on all aspects;</li><li>• To be loyal and trust-worthy to the teachers and others;</li><li>• Help keeping an eco-friendly environment in the campus.</li></ul> |

Prepared by:

Checked by:

Approved by:

**Md. Abdul Hye**

Assistant Professor

Department of Mathematics and statistics

Bangladesh University of

Business and Technology (BUBT)